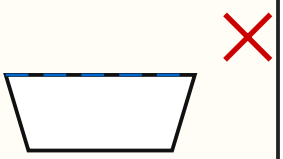
Robotics Project #13 · Templates 5/5 (1:1 scale)

Wheelbarrow Walker Robot

PRINT AT 100% / ACTUAL SIZE
Do NOT 'fit to page'
Check the 100 mm ruler first!

cut (solid) — fold (dashed) — hole: grown-up pokes with a pencil/awl — red = size in mm

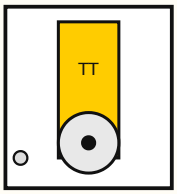
1 Cut and fold the barrow tray (sheet 1). Glue the tabs inside.



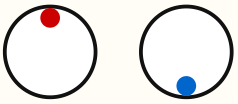
2 Skewer axle through a straw under the tray; glue a wheel on each end. Wheels roll freely.



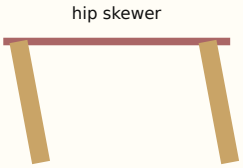
3 Hot-glue a hip plate (sheet 3) on each flat side of the TT motor, motor upright, shaft in the Ø30 hole.



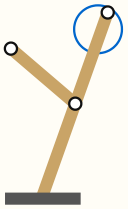
4 Glue crank A on one shaft end, crank B on the other: the pins must point opposite ways.



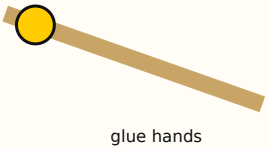
5 Push the 40 mm hip skewer through both hip holes. Put a thigh on each end, add straw stoppers.



6 Leg stick: top hole on the crank pin, middle hole pinned to the thigh knee, foot at the bottom.



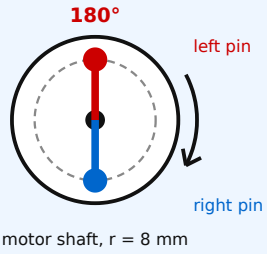
7 Glue torso covers, arms at the shoulders, hands onto the handles with both feet on the table.

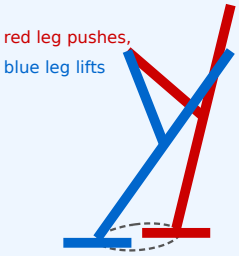


8 2xAA + switch under the tray, wires along a handle to the motor. Switch on and he walks!



How the legs walk





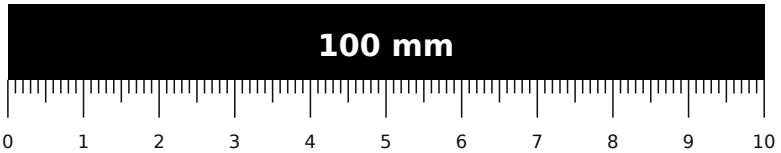
The motor turns both cranks. The pins are half a turn apart, so when the left foot pushes back on the table, the right foot lifts and swings forward - then they swap. Each push is ~22 mm; the foot lifts ~9 mm. The thigh is a guide: it holds the knee so the foot draws a flat-bottomed loop. His hands push the barrow along!

Parts checklist

- ☐ 3 mm cardboard
- ☐ Paper straw
- ☐ Rubber bands (feet)
- ☐ ~10 ice-cream sticks
- ☐ TT motor (double shaft)
- ☐ Ping-pong ball + cap
- ☐ 2 wheels (caps/65 mm)
- ☐ 2xAA holder + batteries
- ☐ Hot glue (grown-up)
- ☐ Skewers 3 mm
- ☐ Slide switch + wire
- ☐ Pencil, scissors

Safety first!

- A grown-up uses the hot glue gun, craft knife, and pokes the holes.
- Snip the sharp skewer tips off.
- Fingers away from the cranks when ON.
- Batteries the right way round; never join the wires; take them out after.
- Keep small parts from little kids.



20 mm

Calibration: the black bar must measure exactly 100 mm and the square 20x20 mm. If not, reprint at 100% (Actual size). Use card ~1 mm (cereal box) for parts.

github.com/dev-repos/kids-robotics-projects · Kids.JoinDev · CC BY 4.0 · Grown-ups do all craft-knife cutting.